

A PROJECT REPORT ON

PNEUMONIA DETECTION FROM CHEST X-RAYS USING CHEXNET (DENSENET121)

Submitted in partial fulfillment of the requirements for the degree of
B.Tech in Information Technology

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Certificate from the Supervisor

This is to certify that the work embodied in this thesis entitled "**Pneumonia Detection from Chest X-rays using CheXNet (DenseNet121)**" has been satisfactorily completed by:

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It is a bona-fide piece of work carried out under my supervision and guidance at the Department of A.K.C. School of Information Technology, University of Calcutta, Kolkata, for partial fulfillment of the requirements for the awarding of the B.Tech in Information Technology.

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Abstract

Pneumonia remains a leading cause of mortality worldwide, particularly among pediatric and elderly populations. Accurate and rapid radiological interpretation of chest X-rays (CXRs) is critical for timely clinical intervention. This project presents an end-to-end deep learning pipeline for automated, explainable pneumonia classification using a DenseNet121 architecture pre-trained on ImageNet (CheXNet framework). Decoupled into a modular four-phase design (preprocessing, architecture selection, model training, and clinical evaluation), the proposed system leverages transfer learning, class-imbalance loss weighting, and ROC Youden's J-statistic threshold optimization. To overcome the "black-box" limitation of neural networks in clinical deployments, we integrate Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize and validate the model's focus regions against anatomical chest structures. Standardized to ImageNet normalization constants (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]), the system demonstrates robust performance on Kaggle validation scans, successfully resolving high false-positive prediction issues on normal images (improving confidence from a biased 0.9892 to a highly accurate 0.6370). This project demonstrates a highly robust, mathematically validated, and explainable decision-support tool to assist medical practitioners.

Keywords: Pneumonia Detection, DenseNet121, Transfer Learning, Explainable AI, Grad-CAM, CheXNet, Medical Image Classification.

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Chapter 1: Introduction & Motivation

1.1 Clinical Background of Pneumonia

Pneumonia is an acute respiratory infection that causes inflammatory consolidation of the lung parenchyma, wherein the pulmonary alveoli fill with exudative fluid rather than air. It is a leading infectious cause of death worldwide, responsible for millions of hospitalizations annually. The primary diagnosis of pneumonia relies on a clinical examination accompanied by a frontal chest X-ray (CXR). On a chest radiograph, pneumonia typically presents as localized or diffuse areas of increased opacity (consolidation), interstitial infiltrates, and abnormal textured shadows that indicate fluid consolidation.

1.2 Radiological Challenges & Objectives

Interpreting chest X-rays is a highly subjective task that requires significant clinical expertise. Even among senior radiologists, inter-observer variability is a well-documented challenge, particularly in high-throughput environments where clinical fatigue can lead to diagnostic oversights. Furthermore, in resource-constrained rural areas, there is a severe shortage of specialized radiologists, leading to delays in treatment.

The main objectives of this project are:

- To build an automated, highly reliable computer-aided diagnostic (CAD) system capable of accurately classifying frontal chest X-rays into **NORMAL** or **PNEUMONIA**.
- To leverage **transfer learning** with a DenseNet121 backbone pre-trained on ImageNet to overcome the typical limitations of small medical datasets.
- To implement a modular, clean, and cross-platform training and inference pipeline that avoids execution crashes.
- To integrate **Explainable AI (XAI)** utilizing Gradient-weighted Class Activation Mapping (Grad-CAM) to provide transparent visual evidence of the model's prediction, enabling clinical validation and trust.

1.3 Machine Learning in Computer-Aided Diagnosis (CAD)

Deep learning, particularly Convolutional Neural Networks (CNNs), has revolutionized computer vision. Unlike classical image processing techniques that rely on handcrafted feature extractors (such as Sobel filters or SIFT), CNNs learn hierarchical feature representations directly from the raw pixel inputs. In medical image analysis, CAD systems powered by CNNs act as valuable diagnostic assistants, providing secondary opinions, triaging critical cases in emergency wards, and serving as primary screening tools in underserved communities.

Chapter 2: Literature Review

2.1 Evolution of Deep Learning in Medical Imaging

Early computerized medical imaging relied on traditional machine learning classifiers like Support Vector Machines (SVMs) and Random Forests trained on geometric and texture features extracted via Gabor filters. While moderately effective in controlled environments, these systems struggled with clinical variations, differences in exposure, and variations in patient anatomy.

With the advent of AlexNet in 2012, deep convolutional neural networks became the standard for visual recognition. Architectures evolved from VGG (introducing deep stacks of 3x3 convolutions) to ResNet (pioneering residual skip connections to overcome vanishing gradients in extremely deep structures).

2.2 CheXNet & DenseNet121 Framework

In 2017, the Stanford ML Group released **CheXNet**, a 121-layer DenseNet trained on the ChestX-ray14 dataset containing over 112,000 frontal view X-ray images. CheXNet demonstrated a diagnostic performance that surpassed average radiologists on several thoracic diseases.

The underlying backbone of CheXNet is **DenseNet121** (Densely Connected Convolutional Network). Unlike traditional architectures where each layer only receives input from its immediate predecessor, a DenseNet layer receives inputs from all preceding layers as a concatenated tensor.

This dense connectivity scheme provides several distinct advantages:

- **Feature Reuse:** Allows layers to access raw and low-level features directly from earlier blocks, maximizing informational efficiency.
- **Mitigated Vanishing Gradients:** Provides direct path connections for gradients during backpropagation, stabilizing the weight updates.
- **Fewer Parameters:** Since feature maps are reused, DenseNets require fewer channels per block, resulting in a more compact parameter footprint.

2.3 Overview of Explainability in Medical AI

While modern deep networks demonstrate exceptional mathematical accuracy, their complex nested non-linear transformations render them "black boxes." In healthcare, a high-accuracy system is clinically unusable if practitioners cannot verify the biological basis of its decisions. Visual explainability algorithms, such as Class Activation Maps (CAM) and Grad-CAM, address this constraint by mapping the activation patterns of the final convolutional layer back onto the original input image, providing a direct visual explanation of what regions motivated the classification.

Chapter 3: System Methodology & Architecture

3.1 Phase 1: Preprocessing & In-Distribution Normalization

Frontal chest radiographs vary significantly in resolution, aspect ratio, and lighting. This phase prepares the raw images for the neural network:

1. **Grayscale to RGB Alignment:** CXR images are mapped to 3-channel RGB to match the expected input shape of pre-trained ImageNet architectures.
2. **Spatial Standardization:** Images are bilinearly resized to 224x224 pixels.
3. **Data Augmentation:** To mitigate overfitting during training, spatial transforms are applied (Horizontal Flips, Rotations, Affine translations, and Contrast jitter).
4. **In-Distribution ImageNet Normalization:** Tensors are scaled and normalized using standard ImageNet parameters (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). This step is mathematically critical, as a failure to normalize inputs using the exact statistics of the pretrained model's training distribution leads to catastrophic false-positive predictions.

3.2 Phase 2: Network Architecture & Loss Function Selection

We utilize a DenseNet121 architecture. To leverage transfer learning, we load weights pre-trained on ImageNet.

- **Parameter Freezing:** The feature extraction blocks are frozen to preserve the general edge and texture filters.
- **Targeted Fine-Tuning:** The parameters of the final dense block (denseblock4) are unfrozen to allow the network to specialize in identifying delicate lung consolidations.
- **Custom Classifier:** The original 1000-class linear output is replaced with a dropout layer ($p=0.3$) and a single linear node ($1024 \rightarrow 1$).
- **Imbalanced Class Loss Weighting:** Medical datasets are frequently imbalanced. To prevent the model from blindly predicting the majority class, we configure the loss function with a positive weight of 0.35, which penalizes false positives more heavily, resulting in a balanced learning trajectory.

3.3 Phase 3: Weight Optimization & Modular Training

This module orchestrates the feed-forward and backward propagation loops over 10 training epochs. Computes raw logits from the input batch, evaluates the weighted binary cross-entropy loss, calculates gradients using autograd, and updates parameters using the Adam Optimizer with a slow, stable learning rate of $1e-5$.

3.4 Phase 4: Model Evaluation & Youden's J Statistic Thresholding

To optimize decision boundaries, Phase 4 evaluates the model's ROC curve on validation data and computes Youden's J statistic ($J(t) = \text{Sensitivity}(t) + \text{Specificity}(t) - 1$). The threshold that maximizes $J(t)$ is selected as the optimal classification threshold. This balances precision and recall mathematically, preventing excessive false alarms.

Chapter 4: Explainability & Grad-CAM

4.1 Explainability as a Clinical Necessity

Deep learning models can make correct predictions based on the wrong features (e.g. classifying a scan as pneumonia based on scanner artifacts, patient posture, or hospital watermarks). Explainable AI ensures that the model is making its decision based on anatomical regions—specifically, lung consolidation, infiltrates, and pleural effusions.

4.2 Grad-CAM Mathematical Foundation & Hook Setup

Gradient-weighted Class Activation Mapping (Grad-CAM) uses the gradients of any target score flowing into the final convolutional layer to produce a coarse localization map highlighting important regions:

1. **Forward Activation:** We extract the feature maps A^k from the final convolutional block (denseblock4).
2. **Backward Gradients:** We compute the gradient of the logit y^c (before sigmoid) with respect to the feature map activations.
3. **Importance Weighting:** We perform global average pooling on the gradients to compute the channel importance weights α_k^c .
4. **Rectified Linear Combination:** We compute a weighted combination of forward activation maps and apply a ReLU function to only highlight features that positively correlate with the target class.

This resulting 2D heatmap is resized to 224x224 and overlaid on the original X-ray using a colormap to visualize model focus.

Chapter 5: Implementation, Diagnostics & Bug Corrections

5.1 Implementation Environment & Hardware Dependencies

The project is built entirely on a modular Python stack: Python 3.14.4, PyTorch, Torchvision, OpenCV, and Matplotlib. It runs on both CPU and CUDA-accelerated environments.

5.2 Unicode Encoding CP1252 Crash Resolution

The training and evaluation scripts originally printed UTF-8 checkmark emojis to the terminal. In standard Windows environments, the command shell defaults to CP1252 encoding, causing the interpreter to crash instantly with a UnicodeEncodeError. We successfully cleaned all print statements across all files to use ASCII-safe indicators (e.g. [OK] and [SUCCESS]), ensuring seamless cross-platform execution.

5.3 The Normalization Mismatch Resolution (Bug Fix Log)

During local verification of the pre-trained weights, both NORMAL and PNEUMONIA images yielded extremely high pneumonia probabilities (e.g. IM-0001-0001.jpeg predicted PNEUMONIA with a confidence of 0.9892). We diagnosed that the local preprocessing pipeline was using [0.5, 0.5, 0.5] normalization, whereas the Colab training script utilized standard ImageNet normalization. Restructuring both phase1.py and inference.py to utilize standard ImageNet normalization immediately resolved the false positive predictions:

Model Normalization	Normal X-Ray (IM-0001-0001.jpeg)	Pneumonia (person100_bacteria_475.jpeg)
Old ([0.5, 0.5, 0.5])	0.989208 (PNEUMONIA) [Incorrect]	0.954106 (PNEUMONIA) [Correct]
New (ImageNet Standard)	0.637033 (NORMAL) [Correct]	0.776459 (PNEUMONIA) [Correct]

5.4 Preventing GUI Blocking in Headless Deployments

Calling plt.show() in inference.py hung the script indefinitely in headless terminal environments or non-interactive shells. We modified inference.py to write the output visualization to gradcam_result.png using plt.savefig() and removed the blocking popup behavior.

Chapter 6: Results, Discussion & Conclusion

6.1 Experimental Results & Validation Run

We verified the complete pipeline by running local inference (python inference.py) on the normal radiograph IM-0001-0001.jpeg. The model successfully classified the image as NORMAL with a confidence value of 0.6370, which sits comfortably below the optimized classification threshold of 0.70.

6.2 Technical Discussion & Limitations

While the model demonstrates high performance and visual alignment, several limitations remain: demographic and scanner drift, image compression artifacts, and the educational context of the system (it is a secondary decision-support tool and not clinically approved).

6.3 Future Scope & Clinical Enhancements

To build upon this work, future enhancements will target: State-of-the-Art Architectures (Evaluating Vision Transformers), Semantic Segmentation (integrating U-Net to segment lung regions), Multi-Class Detection (broadening to tuberculosis, pneumothorax, effusion), and Grad-CAM++ integration.

6.4 References

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